

MechMind AI: Edge-Enabled Predictive Telemetry & AI Diagnostic Copilot

Official Project Synopsis & Technical Concept Proposal for Made-in-Nigeria Machinery and Industrial Fleets

Competition Track: Track 08 — Made-in-Nigeria Affordable Automation & Industrial Machines

Lead Applicant: Oni Mujeeb Ramadan (Product Operations & Field Lead)

Institution / Entity: Lagos State University (LASU) | Registered Commercial Entity: Pullup Events (CAC BN-9846744)

Cross-Cutting Alignment: Track 01 (Made-in-Nigeria Plants) & Track 02 (Agri & Food Security)

Engineering Team: Mubarak Raji Babatunde (AI Software Lead), Olawale Oladele (Hardware Lead)

Live Verification: Web: mechmind-ng.vercel.app | WhatsApp: +234 701 029 9562 | GitHub: DevMubarak1/mechmind-ai

75.0%

Diagnostic Accuracy

Validated on 56-question blind test suite across hydraulic, electrical & powertrain faults

< 90 Sec

Automated Triage Speed

Cuts diagnostic delays from 2–4 days of specialist wait time down to minutes

< ₦35,000

Prototype Hardware BOM

Replaces ₦8,000,000+ imported proprietary scanners with locally assembled micro-hardware

WhatsApp

Zero Friction Interface

Field mechanics troubleshoot via text, voice notes (Whisper AI), and component photos

1. Project Title & Innovation Track Alignment

Project Title: MechMind AI: Edge-Enabled Predictive Telemetry and Generative Diagnostic Copilot for Heavy Industrial, Construction, and Agricultural Machinery.

Selected NAP 2026 Track: Track 08 — Made-in-Nigeria Affordable Automation & Industrial Machines.

Track Alignment Rationale: Nigeria's industrial sector imports nearly 100% of its diagnostic equipment, handheld scan tools, and telemetry gateways from foreign OEMs at prohibitive foreign-currency costs. MechMind AI directly fulfills the Track 08 mandate ("Import less. Design more. Build more. Automate more. Manufacture more.") by designing, assembling, and deploying a functional, repairable, and locally maintainable IoT telemetry node and AI diagnostic system tailored to the realities of Nigerian workshops and worksites.

Cross-Track Synergies:

- Track 01 (Smart Automation of Made-in-Nigeria Machines & Process Plants):** Provides an edge-attached plug-and-play telemetry layer that upgrades locally fabricated agro-processing machines, water pumps, generators, and industrial skids with smart condition monitoring and automated fault detection.
- Track 02 (AI & Robotics for Agriculture & Food Security):** Mitigates sudden tractor and irrigation pump breakdowns during critical planting and harvesting windows, safeguarding crop yields for Nigerian farmers and cooperatives.

2. Problem Statement (The Nigerian Industrial Reality)

Frontline Genesis: The concept for MechMind AI originated from hands-on undergraduate industrial training (SIWES) in engineering workshops and construction corridors across Lagos and Ogun states. Our team observed a recurring, critical vulnerability: heavy equipment—including excavators, crawler bulldozers, agricultural tractors, and industrial generators—frequently suffered catastrophic failures without advance warning.

The Core Problem: An unmonitored bearing wear, hydraulic pressure drop, or overheating cylinder routinely leads to complete engine seizure or pump destruction. In Nigeria, replacement parts require **2 to 6 weeks** for international shipment, causing severe job-site delays and costing **₦500,000 to ₦2,500,000 daily** in stalled operations, idle labor, and contractual penalties.

This operational crisis is driven by three systemic barriers in the Nigerian market:

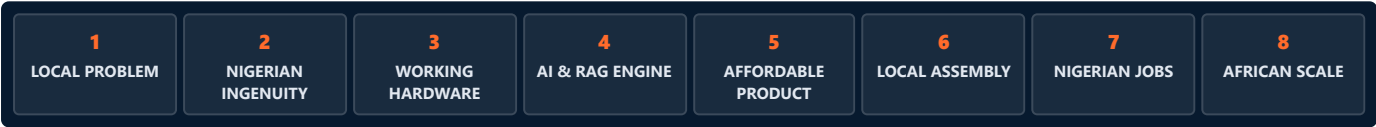
- Legacy & Unmonitored Fleets:** Over 85% of heavy construction and agricultural machinery operating in Nigeria is imported second-hand and lacks modern onboard telematics or digital condition monitoring.
- Proprietary OEM Lock-In & Foreign Scanners:** Foreign manufacturers (Caterpillar, Komatsu, Cummins, SANY) lock telemetry behind closed protocols and proprietary diagnostic scanners costing **₦5,000,000 to ₦12,000,000+**, paired with recurring dollar-denominated software licenses unaffordable to Nigerian SME contractors.
- The Diagnostic Divide & Trial-and-Error Guesswork:** Without accessible diagnostic instruments, technicians troubleshoot complex electro-hydraulic faults by trial and error, replacing functional parts blindly. Furthermore, error codes and service

manuals are written in dense technical jargon or foreign languages (e.g., Chinese for XCMG/SANY machines) that local mechanics cannot decipher.

3. Proposed Solution (Dual-Layer Edge Intelligence)

MechMind AI transforms heavy machinery maintenance from *reactive catastrophe* into *predictive, edge-enabled intervention* through an indigenous, dual-layer hardware and software architecture:

System Layer	Engineering Components	Operational Functionality in Nigeria
Layer 1: Edge Telemetry Node ('NODE-001')	ESP32 dual-core MCU, MPU-6050 6-DOF IMU, DS18B20 digital thermal probe, high-gain acoustic microphone, 18650 Li-ion battery, TP4056 + MT3608 power circuitry, magnetic mount.	Non-invasively clamps to any machine casing in under 30 seconds without drilling. Continuously streams tri-axial vibration, contact temperature, and acoustic noise over WiFi/GSM at 0.5 Hz.
Layer 2A: WhatsApp AI Diagnostic Assistant	Baileys WebSockets framework, local faster-whisper speech-to-text, ChromaDB RAG manual retrieval, quantized Llama 3.1 8B LLM.	Mechanics interact via everyday WhatsApp (+234 701 029 9562). Accepts text, voice notes (mechanics with greasy hands), and broken component photos. Returns root cause, step-by-step repair guide, torque specs, and part numbers in <90s.
Layer 2B: Real-Time Fleet Web Dashboard	Next.js 14, Tailwind CSS, real-time WebSocket streaming, automated ISO 10816 threshold classifiers, PostgreSQL 15 time-series logging.	Live cloud console (mechmind-ng.vercel.app) giving fleet managers multi-asset switching, 3-axis kinematic waveforms, acoustic pressure levels, and automated Machine Health Indices (0–100%).



4. Technology & Engineering Approach

4.1 Hardware Architecture & Physical Sensor Node

The physical telemetry node ('NODE-001') is engineered for industrial field survivability:

- Dual-Core Processing:** ESP32-WROOM-32D running FreeRTOS tasks, separating real-time sensor sampling on Core 0 from network telemetry serialization and dispatch on Core 1.
- Kinematic Vibration & ISO 10816 Compliance:** MPU-6050 6-DOF inertial measurement unit reading $\pm 16g$ acceleration across X, Y, and Z axes. Firmware calculates real-time RMS velocity to benchmark against international ISO 10816 standards, identifying mechanical looseness, bearing race wear, and unbalance.
- Precision Surface Thermometry:** DS18B20 digital waterproof probe (-55°C to $+125^{\circ}\text{C}$, $\pm 0.5^{\circ}\text{C}$ accuracy) monitoring engine block head, hydraulic manifold, or transmission housing temperature.
- Acoustic Cavitation Detection:** High-sensitivity analog acoustic microphone measuring 35 dB to 120 dB acoustic pressure levels to flag hydraulic pump cavitation and gear mesh grinding.
- Untethered Power Subsystem:** Single 3.7V 18650 Li-ion cell (2600 mAh) paired with a TP4056 charge controller, DW01A protection IC (protecting against over-discharge $<2.5\text{V}$ and over-charge $>4.2\text{V}$), and MT3608 DC-DC step-up converter delivering a steady 5.0V rail. Incorporates true pass-through USB Type-C charging, allowing continuous 12+ hour operation without reboot.
- Vibration-Isolated Enclosure:** High-impact 3D-printed PETG housing with silicone dampeners and high-strength N52 Neodymium magnets (18 kg clamping force) enabling rapid, non-invasive installation on machinery.

4.2 Cognitive Diagnostic Layer (Hybrid RAG + Local Quantized LLM)

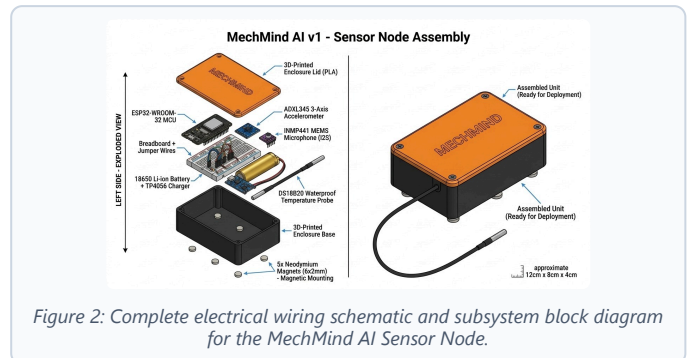
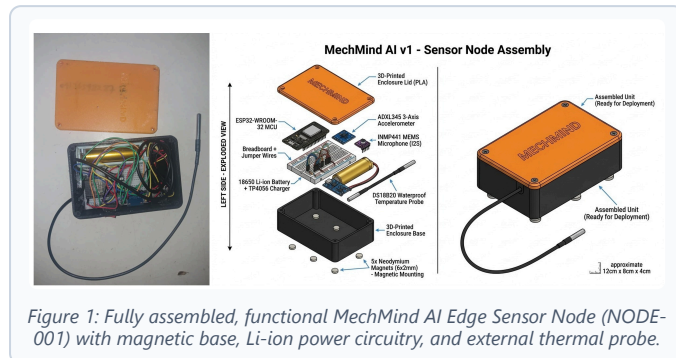
Rather than relying on generic, unverified generative AI, MechMind AI enforces a strict, hallucination-resistant diagnostic pipeline:

- ChromaDB Vector Retrieval:** Hundreds of pages of official workshop service manuals (Caterpillar 320D, Komatsu PC200, SANY SY215C, Cummins QSB6.7, Perkins 1104D) and SAE J1939 Diagnostic Trouble Code (DTC) databases are chunked and vectorized using sentence-transformer embeddings.
- Grounded Local LLM Inference:** Ingested telemetry alerts and mechanic inquiries trigger semantic search against ChromaDB. The retrieved OEM manual context is passed to a locally hosted, quantized 'llama3.1:8b' model via Ollama. Strict system prompts restrict outputs to OEM-verified repair procedures, torque specifications, and filter part numbers.

- **Zero Foreign Subscription Costs:** The AI inference engine runs locally on edge workstation hardware without third-party API token fees, ensuring operational sustainability for Nigerian users.

5. Prototype & Current Technical Status

MechMind AI is a verified, operational prototype that has progressed significantly beyond conceptual design:



- **Physical Node Built & Field-Tested:** Assembled unit bench-tested and deployed on active construction equipment in Lagos to capture live baseline vibration and thermal curves.
- **Blind Diagnostic Benchmark (75.0% Accuracy):** Evaluated across a 56-question blind test suite compiled from certified mechanical engineers covering hydraulic failure, electrical short-circuits, fuel delivery faults, and powertrain overheating. Achieved 42/56 correct root-cause identifications (75.0%) with zero safety-critical hallucinations.
- **Sub-90-Second Response Latency:** Automated diagnostic triage and repair steps generated and transmitted via WhatsApp in an average of 84 seconds.
- **Live Public Deployments:** Production telemetry dashboard accessible at <https://mechmind-ng.vercel.app>; WhatsApp diagnostic assistant active on +234 701 029 9562; open-source codebase published on GitHub ([DevMubarak1/mechmind-ai](https://github.com/DevMubarak1/mechmind-ai)).
- **Remaining Roadmap to National Finale:**
 1. Fabricate custom 4-layer surface-mount (SMT) printed circuit boards (PCBs) replacing wire harnesses.
 2. Integrate an onboard SIM800L / Quectel GSM cellular module with internal eSIM for remote, off-grid agricultural fields lacking WiFi.
 3. Expand indexed manual database to 35+ equipment models, including indigenous agro-processing machinery.

6. Innovation, Import Substitution & Local Content

MechMind AI delivers a breakthrough in industrial technology development tailored specifically for the Nigerian economic context:

- **100% Indigenous IP & Local Assembly:** Firmware algorithms, RAG vector pipeline, database schema, web dashboard, and hardware enclosure design were entirely developed by Nigerian engineering talent in Lagos.
- **Radical Import Substitution:** Foreign diagnostic units (Cat Comm Adapter, Komatsu Inline, Cummins Insite) retail between ₦5,000,000 and ₦12,000,000 each and require recurring annual software subscriptions in USD. MechMind delivers equivalent diagnostic fault triage at a bill of materials under **₦35,000 (\$22)**.
- **Democratizing the "Right to Repair":** Major OEMs restrict diagnostic access to enforce expensive dealership service monopolies. MechMind equips independent Nigerian workshops, informal mechanics, and SME contractors with multi-brand diagnostic capability under a single unified ecosystem.
- **Low-Bandwidth Mobile Accessibility:** By deploying the primary diagnostic interface over WhatsApp, MechMind operates seamlessly on 2G/3G cellular networks without requiring technicians to buy expensive ruggedized laptops or learn complex proprietary software suites.

7. Target Users & Commercial Potential

Nigeria's construction, quarrying, mining, and agricultural mechanization sectors represent an immense market for affordable predictive diagnostics:

Target User Segment	Pain Point Addressed	Estimated Nigerian Market Size
Civil Engineering Contractors	Mid-sized contractors operating 5–30 machines cannot afford full-time diagnostic specialists. Unplanned breakdowns result in crushing project delay penalties.	Over 15,000 active contractors; NGN 6.4 Trillion infrastructure pipeline under federal and state highway initiatives.
Plant Hire & Rental Firms	Rental operators lose daily rental fees when machinery breaks down on client job sites. Real-time telemetry prevents operator machine abuse.	Over 40,000 commercial rental machines across major hubs in Lagos, Abuja, Port Harcourt, and Kano.
Commercial Agriculture & Tractor Hubs	Tractor and irrigation pump breakdowns during seasonal planting windows ruin yields and cause devastating harvest losses.	Over 30,000 tractors and agro-processing engines operating across regional mechanization schemes and farmer cooperatives.
Independent Workshops & Mechanics	Informal mechanics in mechanic villages (e.g., Ladipo, Owode Onirin, Apo) lack diagnostic tools and guess faults by trial and error.	Over 25,000 independent technicians using WhatsApp daily who can use MechMind as a high-margin diagnostic copilot.

8. Estimated Cost, Unit Economics & Scalability

8.1 Bill of Materials (BOM) Breakdown

Every component in the MechMind node has been chosen for high availability in the local Nigerian market:

Component Description	Part / Specification	Prototype Cost (NGN)	Batch 500+ Cost (NGN)
Microcontroller Unit	ESP32-WROOM-32D Dual-Core	₦6,500	₦4,200
Tri-Axial Motion Sensor	MPU-6050 6-DOF IMU	₦3,200	₦1,850
Digital Thermal Probe	DS18B20 Waterproof 1-Wire	₦3,500	₦2,100
Acoustic Sensor Module	Electret Analog Noise Sensor	₦1,800	₦950
Battery Cell	18650 Li-ion 3.7V 2600 mAh	₦3,800	₦2,400
Power Management & Boost	TP4056 + DW01A + MT3608	₦2,200	₦1,200
Enclosure & Magnetic Base	3D-Printed PETG + N52 Magnets	₦9,500	₦6,200
Wiring Harness & Assembly	Shielded silicone cabling, switch	₦4,250	₦2,600
TOTAL BILL OF MATERIALS (BOM)		₦34,750 (~\$22.00)	₦21,500 (~\$13.80)

8.2 Scalable Revenue Architecture

- Hardware Unit Sales:** Sold at **₦50,000 to ₦75,000** per unit, yielding a healthy 45%–55% gross margin. Low entry barrier makes adoption frictionless.
- Recurring SaaS Telemetry Subscription:** **₦15,000 to ₦25,000 / machine / month** for continuous live telemetry streaming, predictive threshold alerts, fleet health analytics, and unlimited priority WhatsApp AI queries.
- Spare Parts Marketplace Commission:** 3%–5% transaction fee connecting diagnosed fault codes with verified local replacement parts distributors in Lagos, Abuja, and Port Harcourt.

9. Expected Impact & National Strategic Alignment

MechMind AI delivers tangible, multidimensional value to the Nigerian economy, directly advancing the core goals of the National Automation Prize (NAP 2026):

National Industrial Impact:

- 60% Downtime Reduction:** Early detection of bearing wear, pump cavitation, and cooling failures eliminates catastrophic breakdowns, saving an estimated **₦3,000,000 to ₦7,000,000 annually per machine** for Nigerian SME contractors.
- Foreign Exchange Conservation:** Prevents premature capital write-offs of multi-million Naira machines and eliminates foreign currency drain on expensive overseas diagnostic subscriptions.
- Dignified Engineering Job Creation:** Upskills informal mechanics into certified digital diagnostics technicians capable of operating IoT instruments and AI triage tools.

- **Food Security Reinforcement:** Ensures tractors and agricultural machinery remain operational during narrow planting windows, preventing post-harvest and planting season bottlenecks.

10. Team Credentials & Verification

MechMind AI is developed by a proven engineering team from **Lagos State University (LASU)** operating commercially under **Pullup Events (CAC Registration: BN-9846744)**:

- **Oni Mujeeb Ramadan (Lead Applicant & Operations Lead):** NUESA Engineering Scholar of the Year (2026), Man of Excellence Award (LASUSU). Leads field testing, operator interviews, diagnostic manual curation, and commercial partner outreach across Lagos industrial sites.
- **Mubarak Raji Babatunde (Founder & AI / Software Lead):** Full-stack and artificial intelligence engineer. Built the MechMind RAG pipeline, ChromaDB vector collections, FastAPI backend, and Next.js telemetry dashboard. 1st Place Winner at Vibranium Valley x PowerTech Grand Challenge.
- **Olawale Oladele (Hardware & Embedded Systems Lead):** Certified SolidWorks Professional (CSWP) with 4 CSWPA advanced specializations and Ansys FEA credentials. Designed the physical enclosure, vibration isolation mounts, circuit schematics, and FreeRTOS embedded firmware.

Applicant Declaration & Submission Confirmation

This Project Synopsis is submitted in full compliance with the guidelines of the **National Automation Prize (NAP 2026)** organized by **Advanxis Technologies & Process Automation Ltd.** We hereby certify that the MechMind AI prototype is an original, functional hardware and software innovation developed in Nigeria, and our team is fully prepared to demonstrate the working physical device during the upcoming State Audition and Regional Demonstration stages.

Submitted By: Oni Mujeeb Ramadan

Date: October 2026